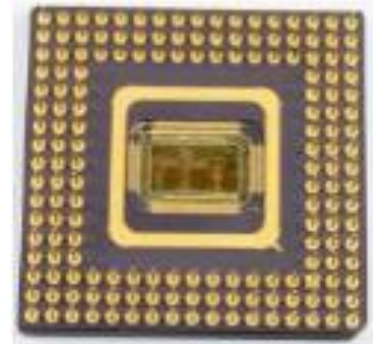


Microprocessors

Microcomputers

There are three main components of a Computer System.

1. **Central Processing Unit (CPU):** Also simply called as the microprocessor acts as the brain coordinating all activities within a computer.
2. **Memory:** The program instructions and data are primarily stored.
3. **Input/output (I/O) Devices:** Allow the computer to input information for processing and then output the results. I/O Devices are also known as computer peripherals.



Microcomputers

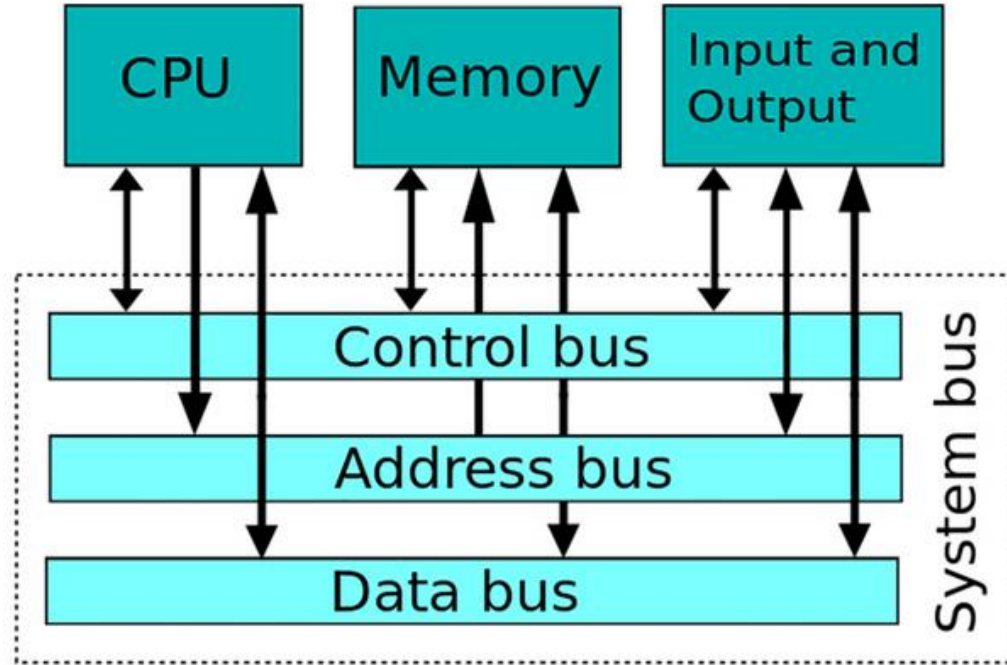
Communicating between these components;

The CPU is connected to memory and I/O devices through a strip of wires called a **bus**.

The bus inside a computer carries information from place to place.

There are three types of busses:

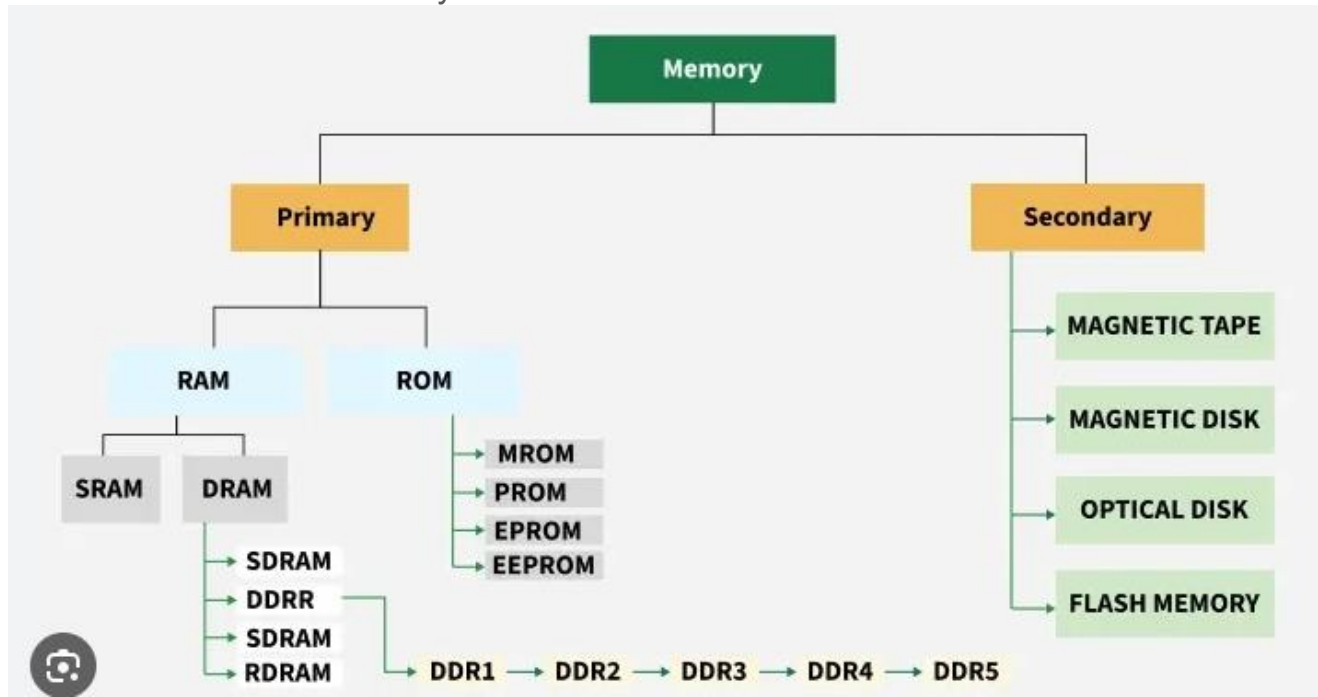
- 1. Address Bus:** is used to identify the memory location or I/O device the processor intends to communicate with. The width of the Address Bus of 8086 microprocessor ranges from **20 bits**
- 2. Data Bus:** is used by the CPU to get data from / to send data to the memory or the I/O devices. The size of data bus of 8086 microprocessor is **16 bits**.
- 3. Control Bus:** How can we tell if the address on the bus is memory address or an I/O device address? This is where the control bus comes in.



Microcomputers

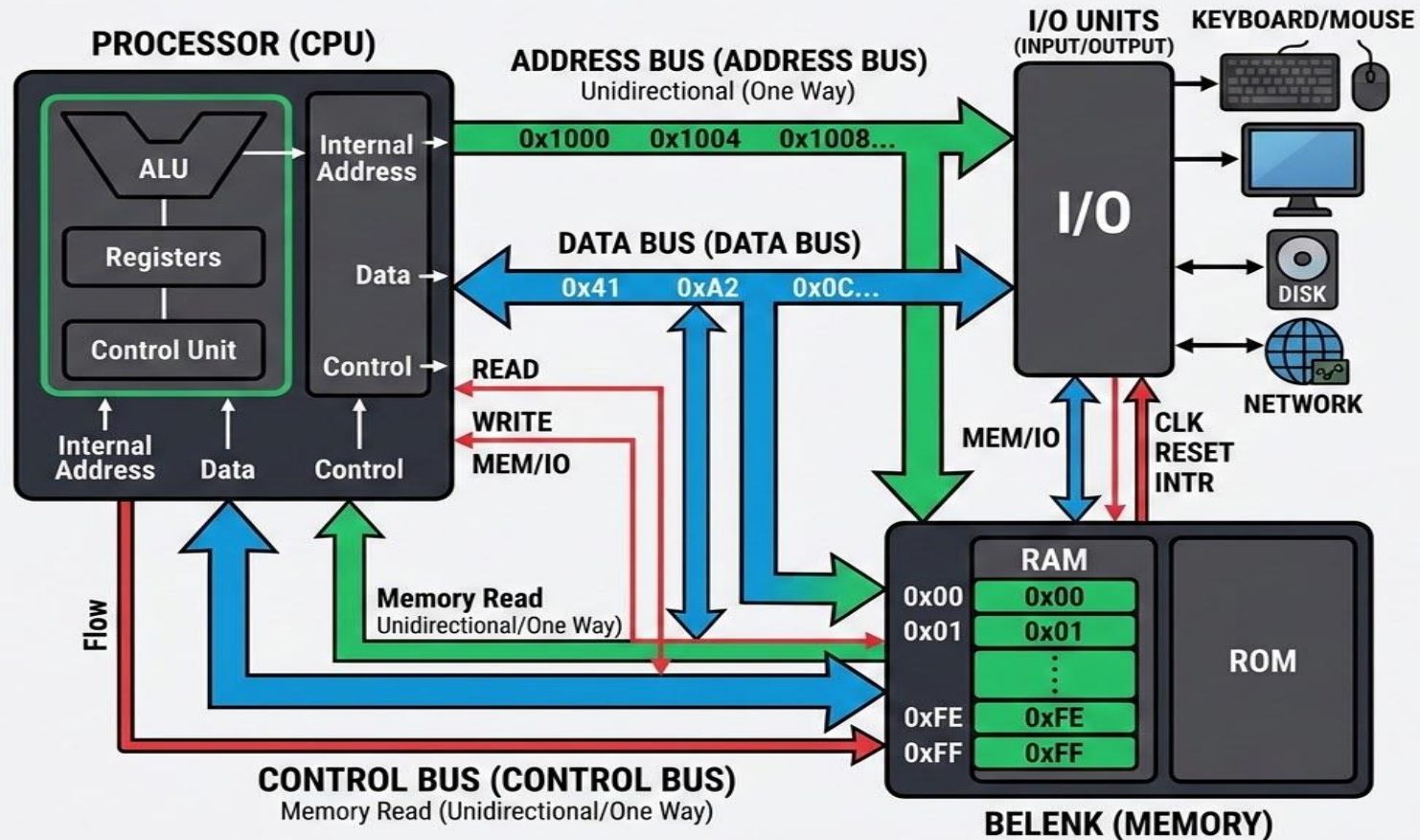
There two types of memory used in microcomputers:

- **RAM (Random Access Memory/ Read-Write memory)** is used by the computer for the temporary storage of the programs that is running. Data is lost when the computer is turned off. So known as **volatile** memory.
- **ROM (Read Only Memory)** the information in ROM is permanent and not lost when the power is turned off. Therefore, it is called **nonvolatile** memory.



SYSTEM BUS ARCHITECTURE (SYSTEM BUS ARCHITECTURE)

CPU-MEMORY AND I/O INTERCONNECTIONS



ENIAC COMPUTER: TECHNICAL SPECIFICATIONS

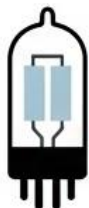
Brief History of the Computers

COMPLETION DATE

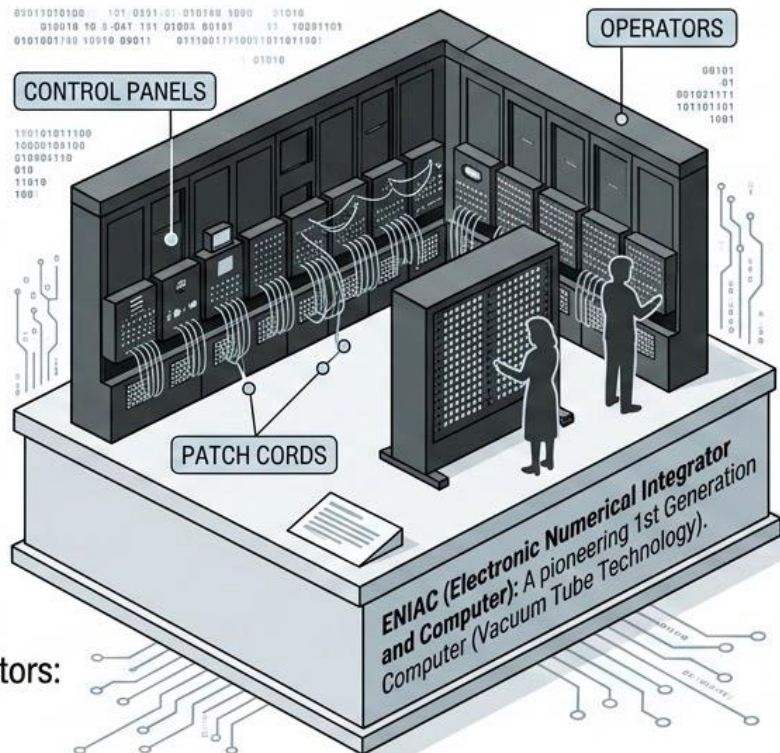


1946 (Development Cycle: 1943–1946)
Moore School of Electrical Engineering

HARDWARE COMPONENTS



Vacuum Tubes: ~17,500
Crystal Diodes: 7,500
Relays: 1,500
Resistors, Capacitors, Inductors: Hundreds of thousands

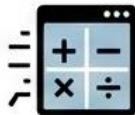


SCALE & PHYSICAL DIMENSIONS



Total Space: ~1,800 sq. feet (167 m²)
Total Weight: ~30 tons
U-Shaped Layout

CAPABILITIES



5,000 Additions / second
357 Multiplications / second
38 Divisions / second

Brief History of the Computers

TRADIC COMPUTER: KEY TECHNICAL SPECIFICATIONS



COMPLETION DATE & DEVELOPMENT

Announced: 1958
Developer: IBM



SCALE & PHYSICAL DIMENSIONS

Floor Space: ~30 sq. feet
Cabinet Weight: ~800 lbs



SOLID-STATE HARDWARE

~800 Transistors (Approx.)
First Fully Transistorized Design

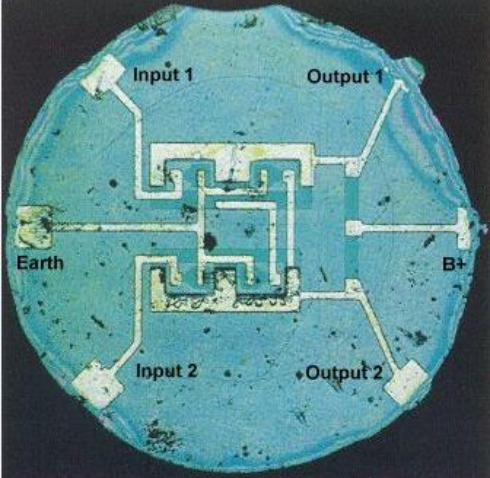


OPERATIONAL CAPABILITIES

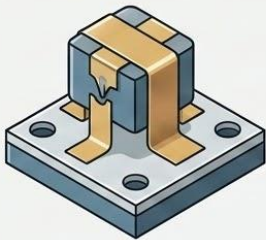
Performance: Arithmetic & Logical Operations in microseconds
Reliability: High-uptime system



Brief History of the Computers

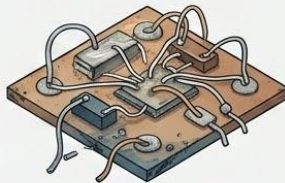


FROM THEORY TO SILICON: KEY MILESTONES TO THE FIRST INTEGRATED CIRCUIT



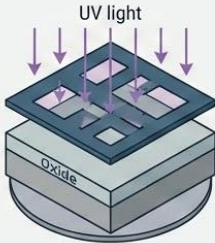
1947
THE TRANSISTOR IS INVENTED
(Bell Labs)

Bell Labs engineers (Shockley, Bardeen, Brattain) replace the bulky vacuum tube with the solid-state transistor, enabling electronics to scale down.



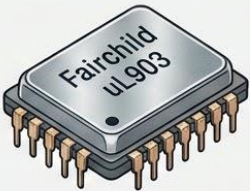
1958
KILBY'S INTEGRATED CIRCUIT
(Texas Instruments)

Jack Kilby demonstrates the first working integrated circuit, showing all components can coexist on a single piece of semiconductor material.



1959
THE PLANAR PROCESSOR
(Fairchild Semiconductor)

Jean Hoerni invents the planar process, a manufacturing method using photolithography to protect the semiconductor surface and make ICs commercially viable.

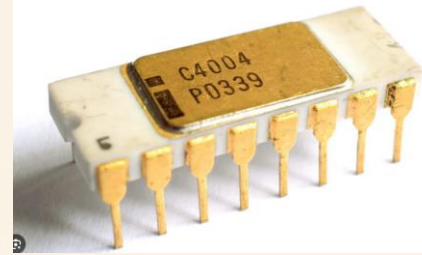


1959
THE FIRST COMMERCIALY VIABLE SILICON IC
(Fairchild Semiconductor)

Robert Noyce combines Hoerni's planar process and metal connections to create the first practical silicon IC, paving the way for mass production.

Brief History of the Computers

1971: The Intel 4004 – Genesis of the Microprocessor

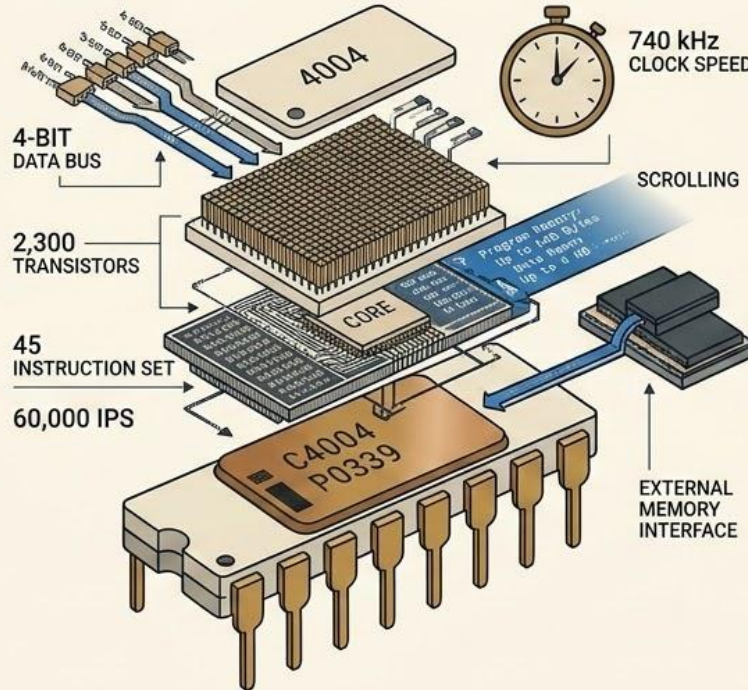


CORE ARCHITECTURE

4-bit
Data Bus
2,300
Transistors

INSTRUCTION PROCESSING

45
Instructions Set
60,000
Instructions Per Second (IPS)
EXECUTION CORE



CLOCK PERFORMANCE

740 kHz
Clock Speed

MEMORY INTERFACE

CPU
Program Memory:
Up to 640 Bytes
Data Memory:
Up to 4 KB
(External Data Memory)

*Data Memory was not on-chip; the 4004 interfaced with external memory for data storage.

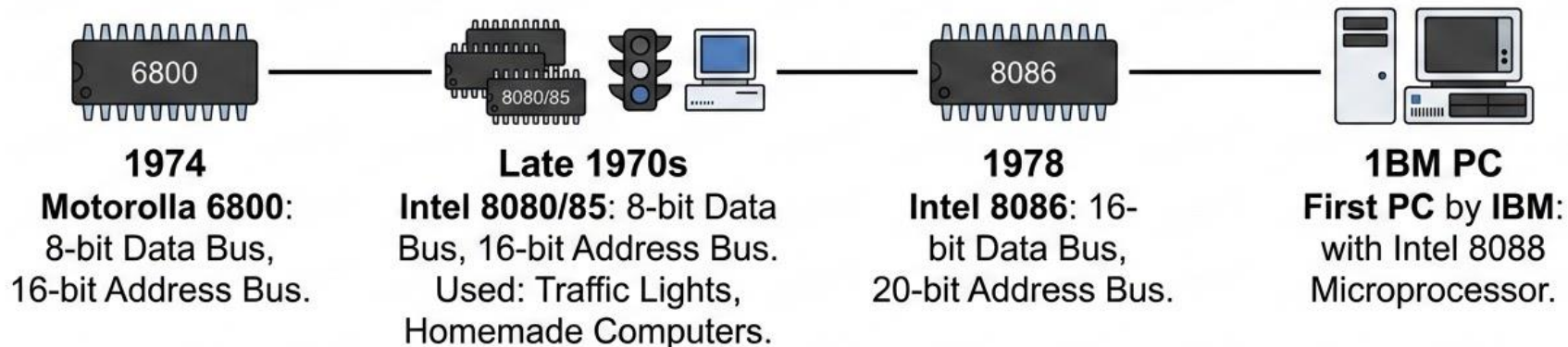
Brief History of the Computers

1974: Motorola 6800 is introduced with 8-bit data bus and 16-bit address bus.

Late 1970s: Intel 8080/85 appeared with 8-bit data bus and 16-bit address bus and used from traffic light controllers to homemade computers.

1978: Intel 8086 is produced with 16-bit data bus and 20-bit address bus.

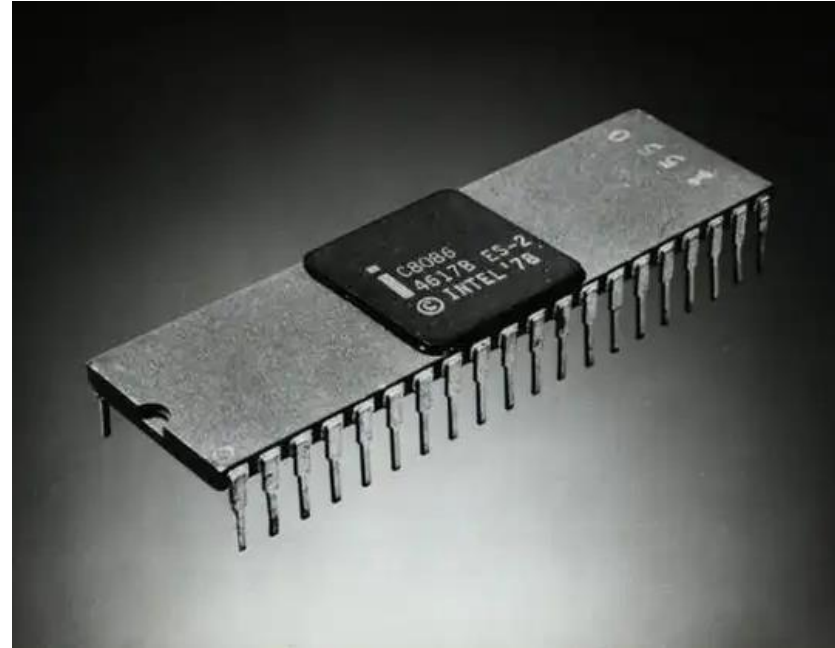
1981: First **PC** was introduced by IBM with **Intel 8088** microprocessor.



Brief History of the Computers

intel 8086

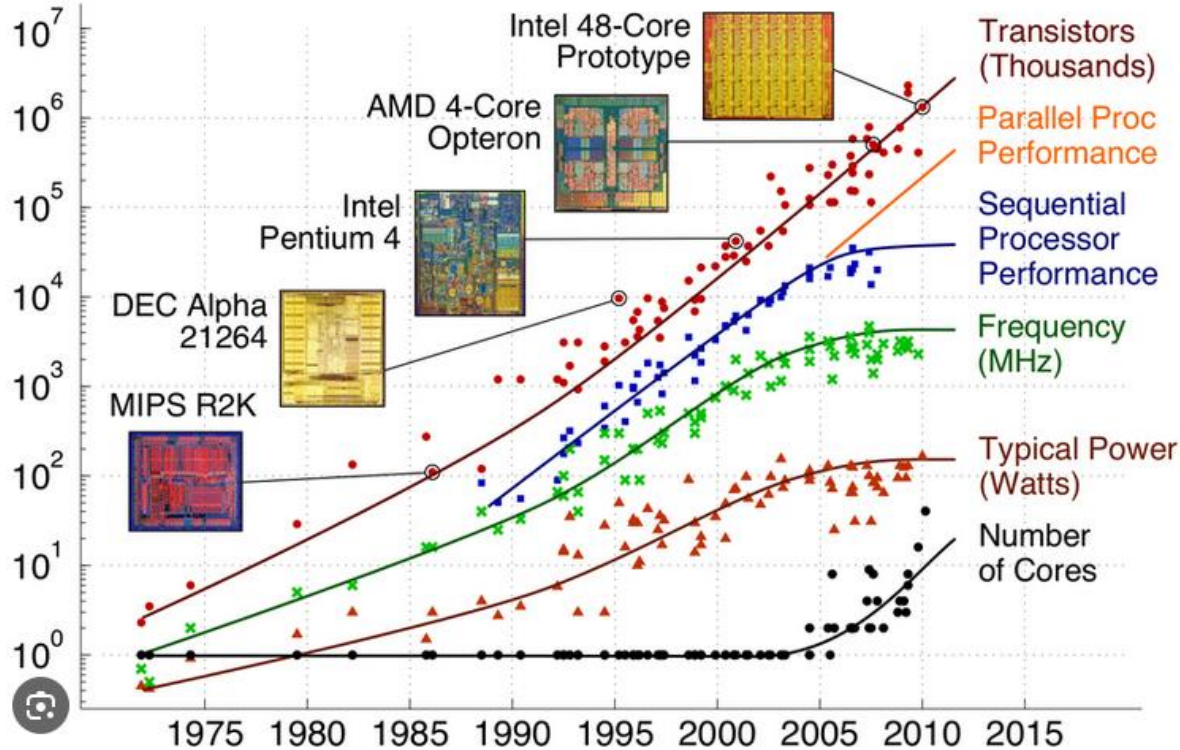
- 16-bit microprocessor.
- Clock speeds ranged from 5 MHz to 10 MHz.
- It could address up to 1 megabyte of memory.
- 20-bit address bus.
- supported a variety of data types and operations, including operations on strings.
- The 8086 introduced a form of **pipelining**, a technique that allows the overlap of instruction fetch and execution, improving the efficiency of instruction processing.



Evolution of Intel 80x86 Family Microprocessors

Processor	Year Intro.	Transistors	Clock Rate (MHz.)	External Data Bus	Internal Data Bus	Add. Bus
4004	1971	2,250	0.108	4	8	12
8008	1972	3,500	0.200	8	8	14
8080	1974	6,000	3	8	8	16
8085	1976	6,000	6	8	8	16
8086	1978	29,000	10	16	16	20
8088	1979	29,000	10	8	16	20
80286	1982	134,000	12.5	16	16	25
80386DX	1985	275,000	33	32	32	32
80386SX	1988	275,000	33	16	32	24
Pentium C	1993	3,100,000	66 -200	64	32	32
Pentium MMX	1997	4,500,000	300	64	32	32
Pentium Pro	1995	5,500,000	200	64	32	36
Pentium II	1997	7,500,000	233-450	64	32	36
Pentium III	1999	9,500,000	550-733	64	32	36
Itanium	2001	30,000,000	800-...	128	64	64

Evolution of Microprocessors



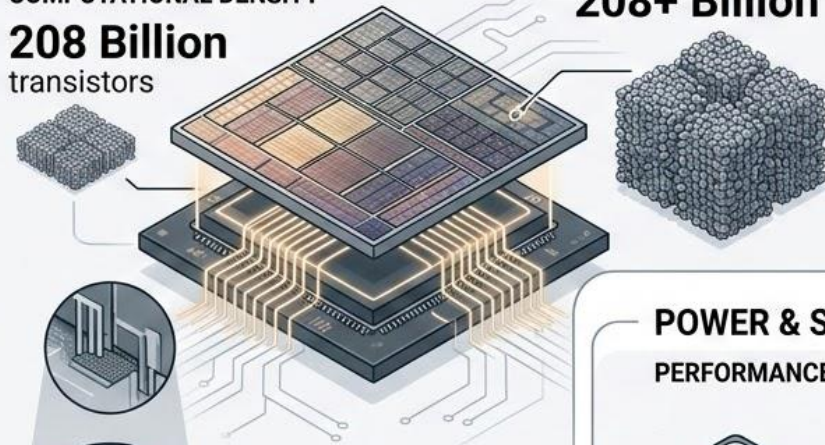
The Modern Microprocessor: Striking Data & Trends

DENSITY & SCALE

COMPUTATIONAL DENSITY

208 Billion
transistors

208+ Billion

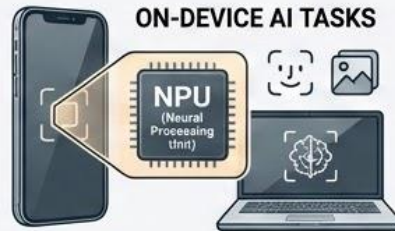


PROCESS NODE: 2nm.
~40,000x smaller
than a human hair.

TRANSISTOR COUNT:
208+ Billion.

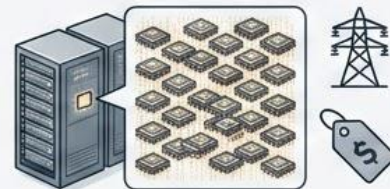
AI HARDWARE INTEGRATION

ON-DEVICE AI TASKS



UBIQUITOUS AI: Dedicated NPUs.

DATA CENTER DEMAND



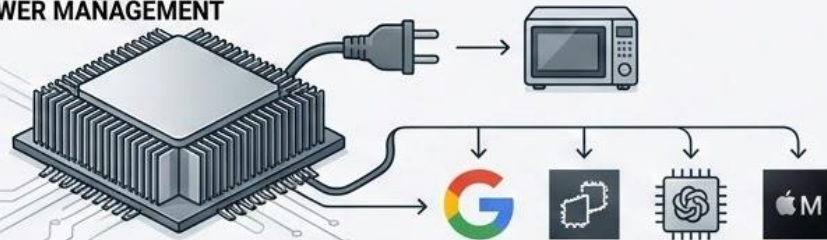
AI TRAINING: Thousands of GPUs.

POWER & SPECIALIZATION

PERFORMANCE & POWER MANAGEMENT



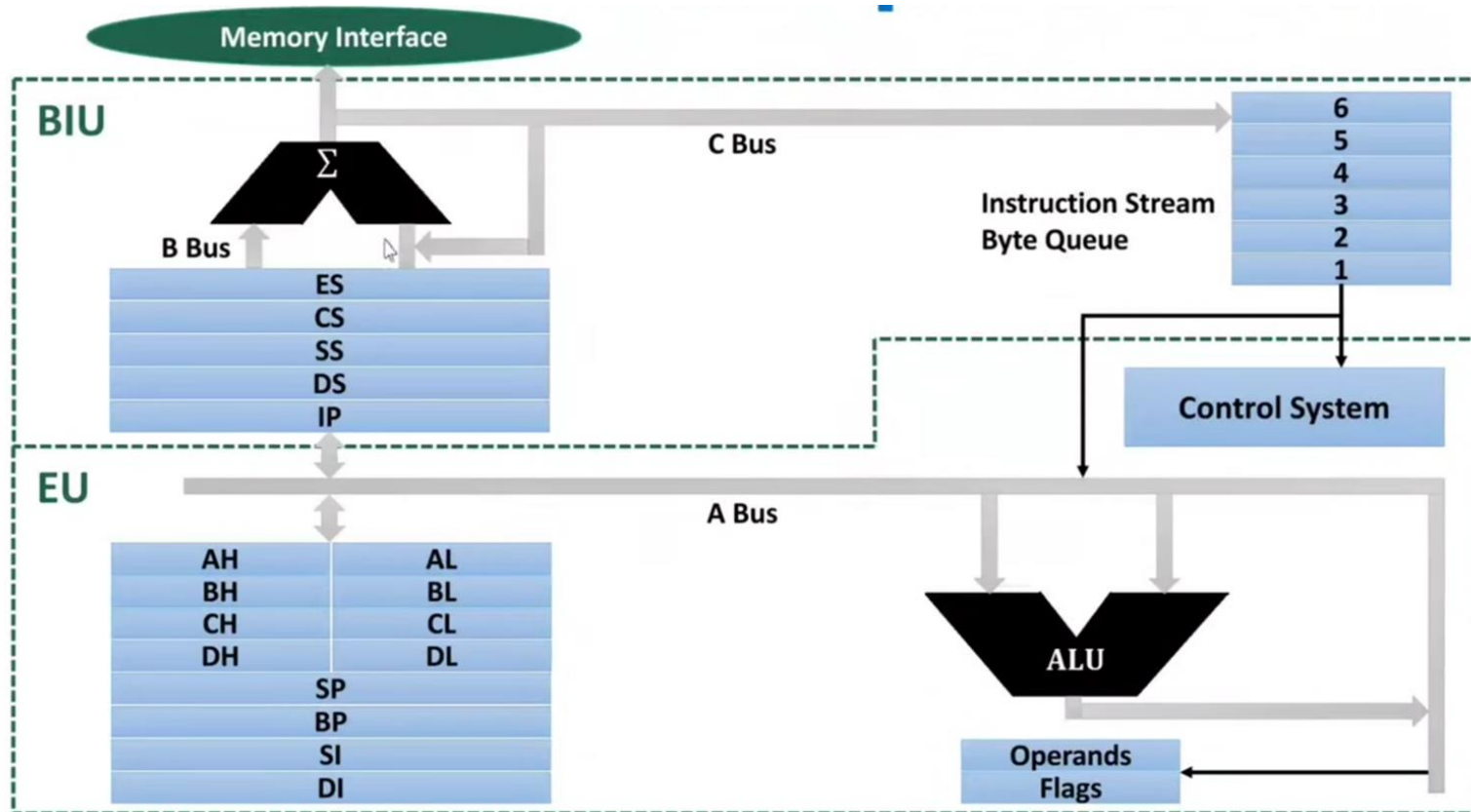
<5W



>1000W (1kW)

SPECIALIZATION IS KEY:
Special-purpose ASICs for specific tasks.

Architecture of 8086 Microprocessor



Bus interface unit (BIU) of 8086

❖ BIU is responsible for establishing communications with external peripheral devices and memory via system bus.

❖ Main Purpose of BIU serves is as followed

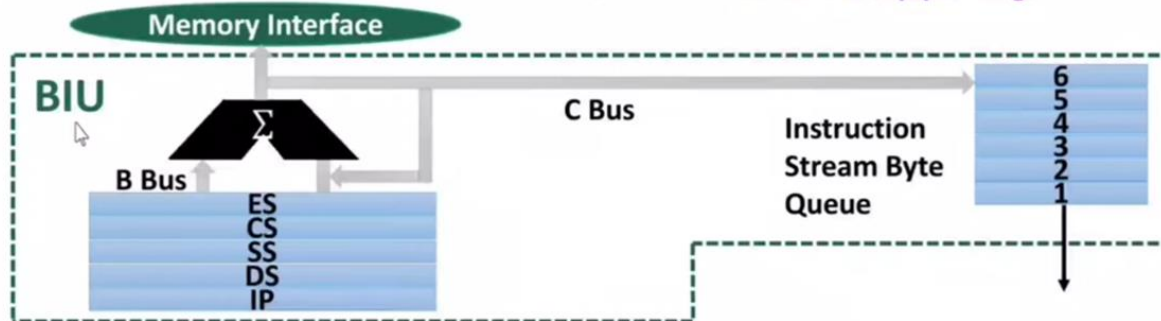
- ❑ It fetches the instructions from Memory
- ❑ It reads data from IO and Memory
- ❑ It writes data into IO and Memory
- ❑ It provides the address relocation facility

❖ BIU contains three main parts

- ❑ Segment Registers
- ❑ Instruction Queue
- ❑ Instruction Pointer

❖ Instruction Queue

- ❑ BIU prefetches six instruction bytes in advance from memory.
- ❑ The prefetched instructions are stored in a group of high speed registers known instruction queue.
- ❑ This Instruction queue works on FIFO order.
- ❑ BIU and EU works in parallel.
- ❑ The simultaneous operations of BIU and EU is possible only when the EU does not require the system bus.
- ❑ The process of fetching the next instruction in advance while the EU is executing the current instruction is pipelining.



Bus interface unit (BIU) of 8086

❖ Segment Register of BIU

- ❑ The 8086 MP has the capability of addressing 1MB memory, which is divided into 16 local segments.
- ❑ Each segment contains 64KB memory
- ❑ But any instant 8086 works with only four 64KB segments.

❖ Each segment associated with segment register

- ❑ Code Segment Register CS – 16bits
- ❑ Data Segment Register DS – 16bits
- ❑ Stack Segment Register SS – 16bits
- ❑ Extra Segment Register ES – 16bits

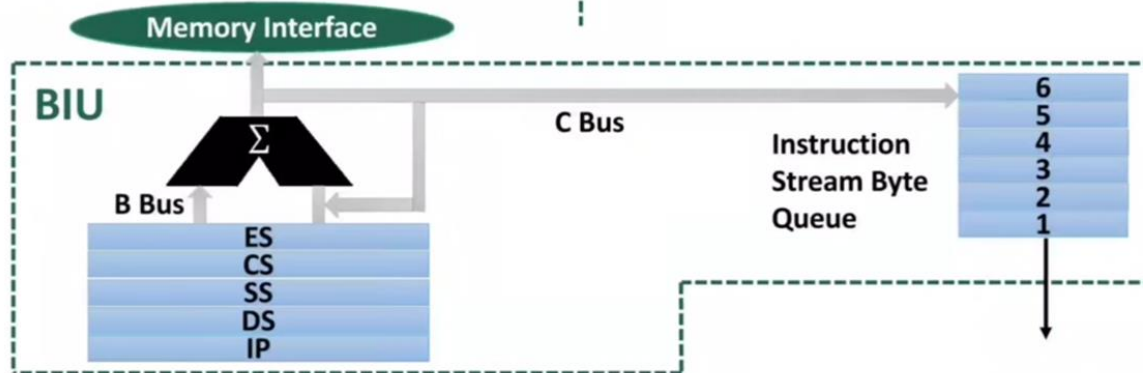
❖ Segment Register of BIU is used to store the starting address of Memory segment.

❖ BIU generates 20 bits address using segment register and Offset pointer registers.

$$PA = SR \times 10H + OP$$

❖ Instruction Pointer of BIU

- ❑ IP holds the address of next instruction which is to be executed next.
- ❑ It contains the OFFSET value of next Address.



Execution unit (EU) of 8086

❖ EU informs BIU from where the next instruction or data to be fetched.

❖ The EU performs following functions:

- ❑ It picks up the instruction from the instruction queue of BIU.
- ❑ It decodes the instructions and then executes the instruction.
- ❑ It updates the status of flag register.

❖ Control Unit

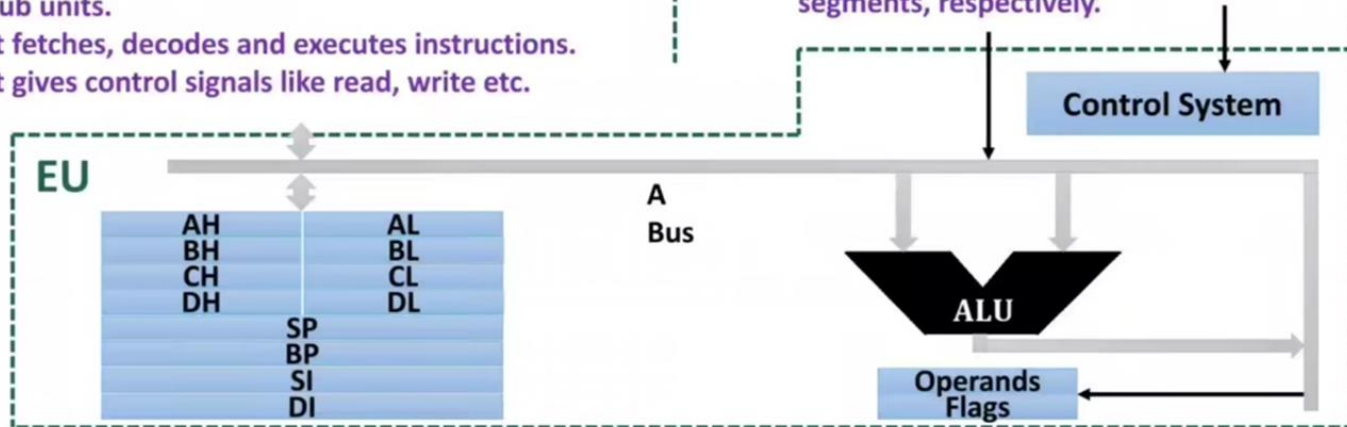
- ❑ It is controlling and coordinating all the activities of sub units.
- ❑ It fetches, decodes and executes instructions.
- ❑ It gives control signals like read, write etc.

❖ ALU of EU

- ❑ It performs all the arithmetic and logical operations.
- ❑ Results may be stored in general purpose registers or index registers.
- ❑ It updates Flag register after every instructions.

❖ General Purpose Registers & Pointer of EU

- ❑ EU have four general purpose registers.
- ❑ These registers are used for accessing data very fast.
- ❑ EU have SP and BP for stack.
- ❑ Index registers are SI and DI for data and Extra segments, respectively.



Pipelining in 8086

❖ How to increase Speed of execution of programs in microprocessor :

[Bu video dosyasını indir](#)

- ❑ Increasing data bus. [So, in single machine cycle we can transfer more data]
- ❑ Increasing System Clock. [So, more cycles can get executed in given time]
- ❑ Upgrading ALU. [Simplest explanation is 16 bits ALU performs more arithmetic and logical operations compared to 8 bits of ALU]
- ❑ Upgrading Memory technology. [Like now a days (2022), we have SSD & Cache in our computer]
- ❑ Upgrading Instruction Set. [Like in 8086 we have Multiplication and Division Instructions available which wasn't available with 8085]
- ❑ **Pipelining [It helps to execute multiple instructions in parallel.]**

8085
Instruction
Execution



8086
Instruction
Execution



The registers of the 8086

General purpose registers

- ❖ **AX (accumulator)** is used for instructions such as multiplication, division, and some of the adjustment instructions. Generally considered to be a multipurpose register.
- ❖ **BX (base addressing register)** holds the offset address of a location in the memory system.
- ❖ **CX (counter loop operations)** holds the count for various instructions.
- ❖ **DX (data in I/O operations)** holds a part of the result from a multiplication or part of the dividend before a division.

General registers can be accessed as full **16 bits** (such as AX), or as the **high byte only** (AH) or **low byte only** (AL). The others are not!!

AX	AH [8]	AL [8]	Accumulator
BX	BH [8]	BL [8]	Base
CX	CH [8]	CL [8]	Counter
DX	DH [8]	DL [8]	Data

General Purpose Registers

Category	Bits	Register Names
General	16	AX, BX, CX, DX
	8	AH, AL, BH, BL, CH, CL, DH, DL
Pointer	16	SP (stack pointer), BP (base pointer)
Index	16	SI (source index), DI (destination index)
Segment	16	CS (code segment), DS (data segment) SS (stack segment), ES (extra segment)
Instruction	16	IP (instruction pointer)
Flag	16	FR (flag register)

The registers of the 8086

Index registers

- ❖ **BP**, points to a memory location for memory data transfers.
- ❖ **DI**, addresses string destination data for the string instructions.
- ❖ **SI**, addresses source string data for the string instructions.
- ❖ **IP**, points to the next instruction in a program, is used by the microprocessor to find the next sequential instruction in a program located within the code segment.
- ❖ **SP**, addresses an area of memory called the stack.

SP [16]	Stack Pointer
BP [16]	Base Pointer
SI [16]	Source Index
DI [16]	Destination Index
IP [16]	Instruction Pointer

Segment Offset Pointers

Category	Bits	Register Names
General	16	AX, BX, CX, DX
	8	AH, AL, BH, BL, CH, CL, DH, DL
Pointer	16	SP (stack pointer), BP (base pointer)
Index	16	SI (source index), DI (destination index)
Segment	16	CS (code segment), DS (data segment) SS (stack segment), ES (extra segment)
Instruction	16	IP (instruction pointer)
Flag	16	FR (flag register)

The programming model of the 8086

Segment registers

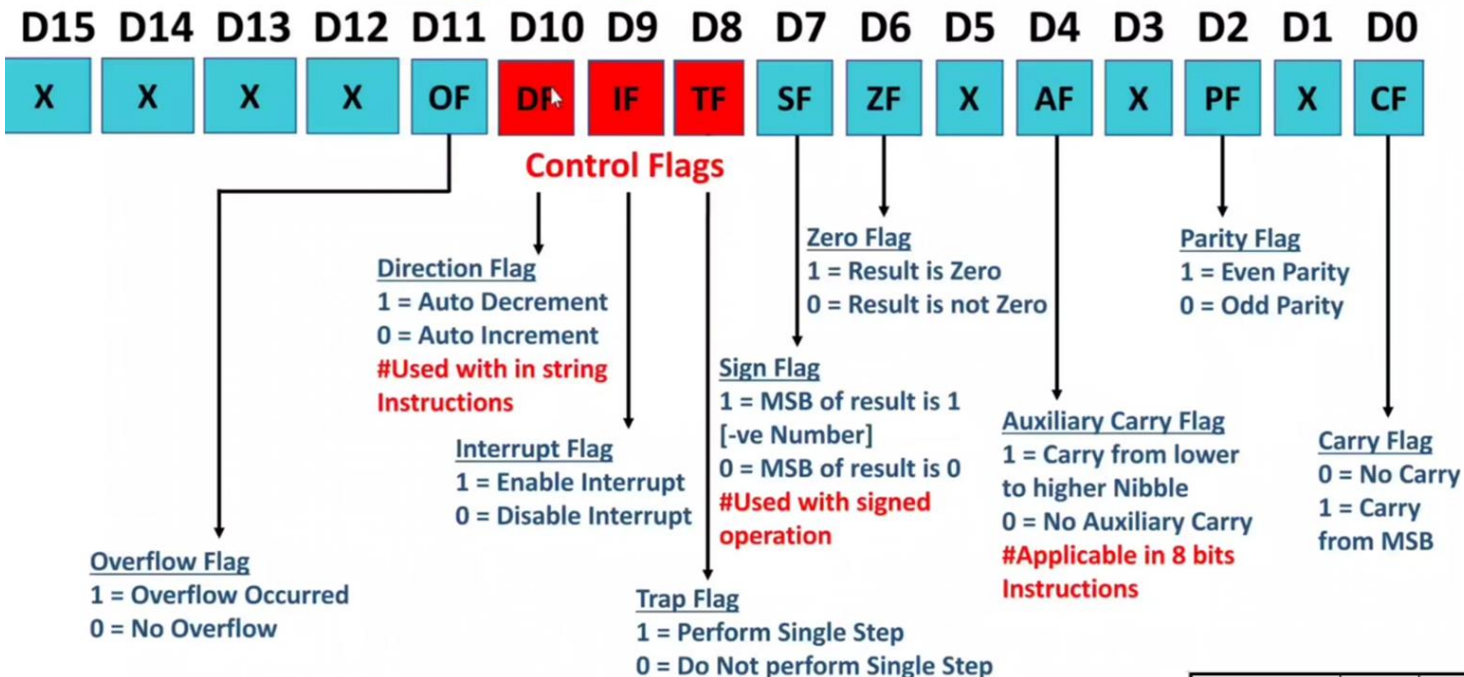
- ❖ **CS**: contains the program code (instructions)
- ❖ **DS**; used to store data to be processed by the program
- ❖ **SS**; used to store information temporarily
- ❖ **ES**; used to store data like strings

ES [16]	Extra Segment
CS [16]	Code Segment
DS [16]	Data Segment
SS [16]	Stack Segment

Segment Registers

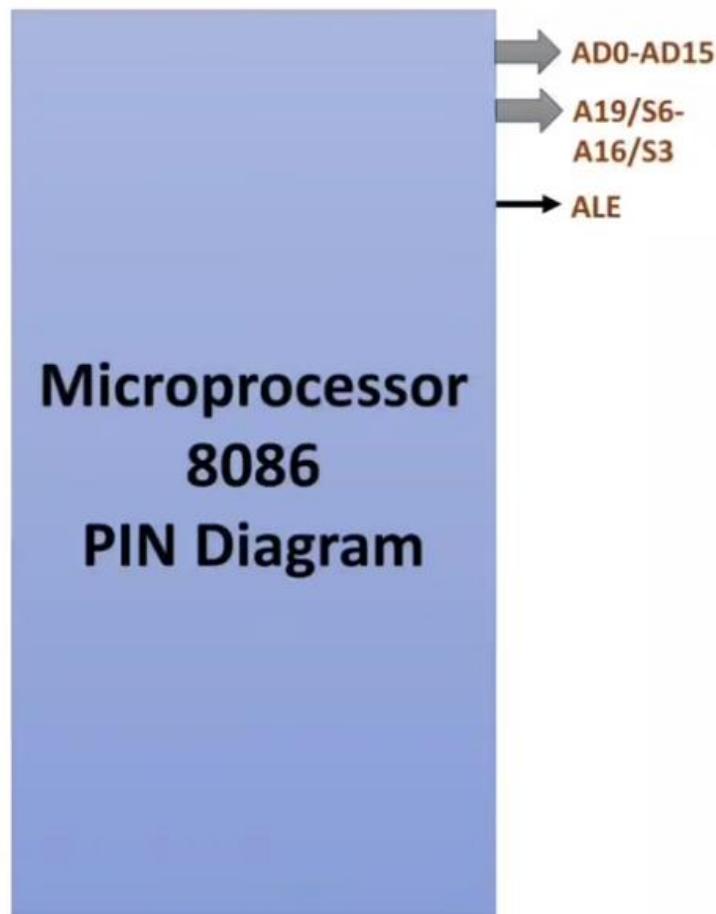
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	8	AH, AL, BH, BL, CH, CL, DH, DL
Pointer	16	SP (stack pointer), BP (base pointer)
Index	16	SI (source index), DI (destination index)
Segment	16	CS (code segment), DS (data segment) SS (stack segment), ES (extra segment)
Instruction	16	IP (instruction pointer)
Flag	16	FR (flag register)

Flag Register of 8086



Category	Bits	Register Names
General	16	AX, BX, CX, DX
	8	AH, AL, BH, BL, CH, CL, DH, DL
Pointer	16	SP (stack pointer), BP (base pointer)
Index	16	SI (source index), DI (destination index)
Segment	16	CS (code segment), DS (data segment)
		SS (stack segment), ES (extra segment)
Instruction	16	IP (instruction pointer)
Flag	16	FR (flag register)

Pin Diagram of Microprocessor 8086



❖ Address Data Bus – [AD0-AD15]

- ❑ 8086 has 20 lines for Address bus and 16 Data Lines.
- ❑ Here, AD0 – AD15 are time multiplexed Address Data Lines and can be separated by ALE terminal.
- ❑ If ALE = 1 then it carries Address [A0-A15] and If ALE = 0 then it carries data [D0-D15].

❖ Address Status Lines – [A19/S6-A16/S3]

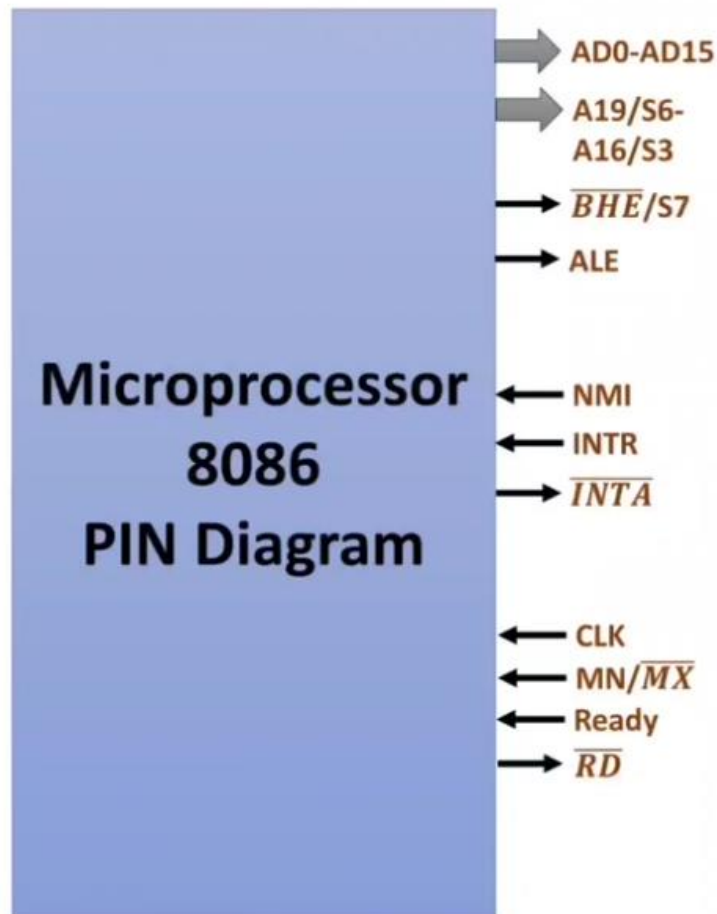
- ❑ 8086 has 4 time multiplexed Address status lines A19/S6 – A16/S3 and can be separated by ALE terminal.
- ❑ If ALE = 1, then it carries Address [A19-A16] and If ALE = 0 then it carries Status signals [S6-S3].

❖ Status signals [S7/S3]

- ❑ S3 & S4 indicates which segment is accessed by 8086 during current bus cycle.
- ❑ S5 reflects IF flag register.
- ❑ S6 is always zero.
- ❑ S7 is always one.

S4	S3	Segment Accessed
0	0	ES
0	1	SS
1	0	CS
1	1	DS

Pin Diagram of Microprocessor 8086



❖ System Clock – [CLK]

- ☐ Clock is given to 8086 for internal timings.
- ☐ Clock with 8086 is 5MHz, 8MHz & 10MHz.

❖ Minimum & Maximum Mode [$\overline{MN}/\overline{MX}$]

- ☐ For Minimum Mode connected with Vcc.
- ☐ With Minimum Mode 8086 will work as single microprocessor.
- ☐ For Maximum Mode connected with ground.
- ☐ With Maximum mode 8086 will work with multiple coprocessor.

❖ Ready Signal [READY]

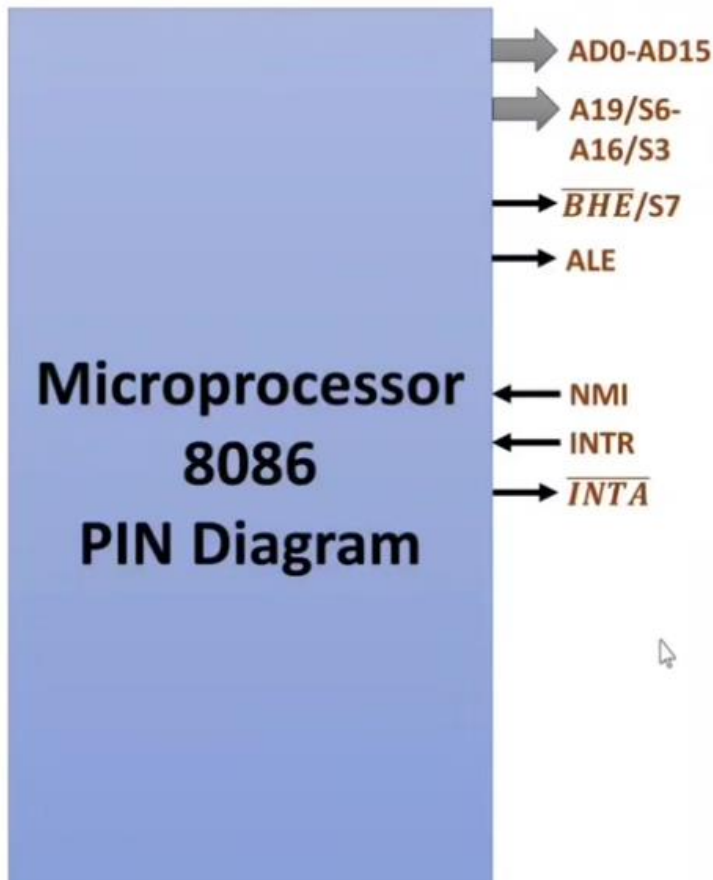
- ☐ This input is used to insert the wait state into timing cycle of microprocessor 8086.
- ☐ If the ready pin is at logic 1, it has no effect on the operation.
- ☐ If it is logic 0, 8086 enters into the wait state like idle.
- ☐ It is used to synchronize slow peripheral devices.

❖ Read [$\overline{RD}$]

- ☐ If it is logic 0, Microprocessor reads data from Memory or IO devices.



Pin Diagram of Microprocessor 8086



❖ Bus High Enable – [$\overline{BHE}/S7$]

- ☐ This line is used to enable D15-D8 of D15-D0.
- ☐ It is used by Microprocessor for memory banking.

❖ Non Maskable Interrupt – [NMI]

- ☐ This line is used give highest priority interrupt to Microprocessor 8086.
- ☐ It can not be disabled by the software.
- ☐ It is positive edge triggered interrupt.
- ☐ When it occurs, Type 2 Interrupt occurs in the 8086.

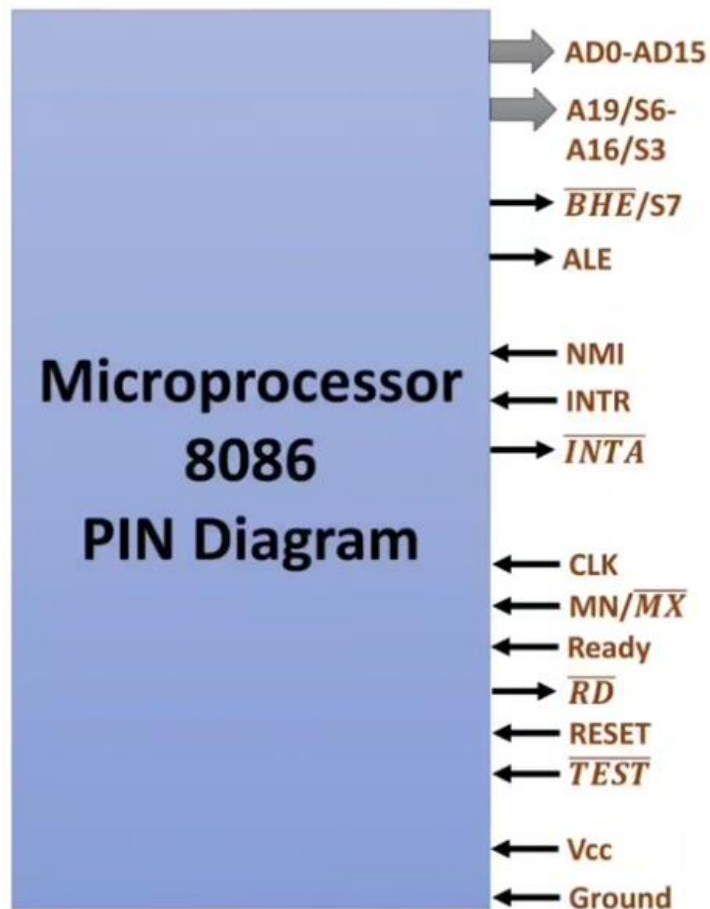
❖ Interrupt Request – [INTR]

- ☐ This is level triggered hardware interrupt.
- ☐ It depends of status of Interrupt Flag.
- ☐ If IF = 1, INTR = 1, the 8086 gets interrupted.
- ☐ If IF = 0, INTR = 1, then INTR is disabled.

❖ Interrupt Acknowledgment – [$\overline{INTA}$]

- ☐ Along with Interrupt request on INTR, Microprocessor gives interrupt acknowledgment to peripherals.

Pin Diagram of Microprocessor 8086



❖ System Reset – [RESET]

- ☐ This input will reset microprocessor 8086.
- ☐ If it held logic 1 for minimum of 4 clock cycles then microprocessor will get RESET.
- ☐ After RESET of 8086, CS and IP initialized to FFFFH and 0000H, so physical address will be FFFF0H.
- ☐ Remaining registers initialized to zero.

❖ Test [$\overline{TEST}$]

- ☐ It is used for synchronization.
- ☐ When this input is logic 0, 8086 executes WAIT instruction.

❖ Power Supply [Vcc & Ground]

- ☐ 8086 is given with 5 Volt DC supply.
- ☐ Allowed variation is 10%.
- ☐ 8086 has two ground pins.
- ☐ This two ground pins are there to have less power dissipation.

Multiplexed Address/Data Bus (AD0-AD15)

Multiplexed Address/Status Bus (A19/S6-A16/S3)

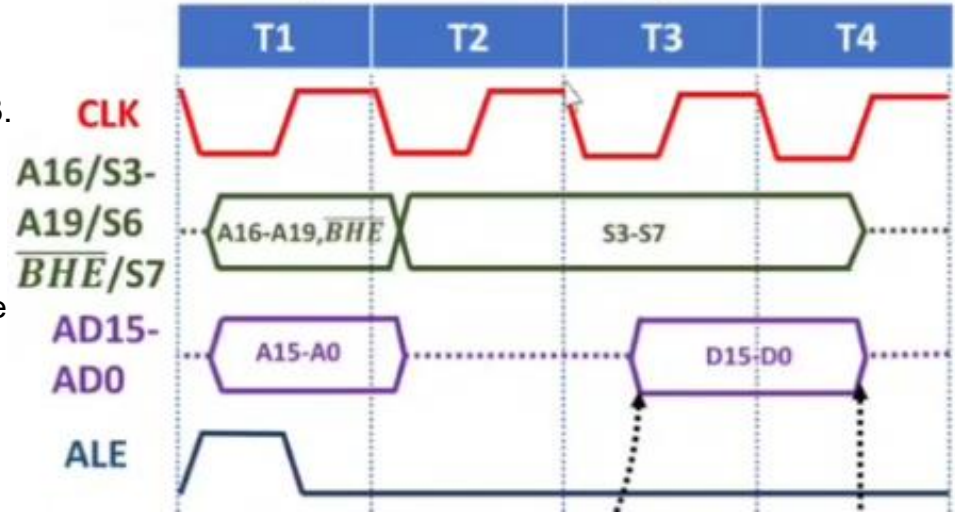
- **T1:** The lines carry the address bits A19-A16.
- **T2, T3, T4:** The lines carry the status signals S6-S3.

Multiplexed Address/Data Bus (AD0-AD15)

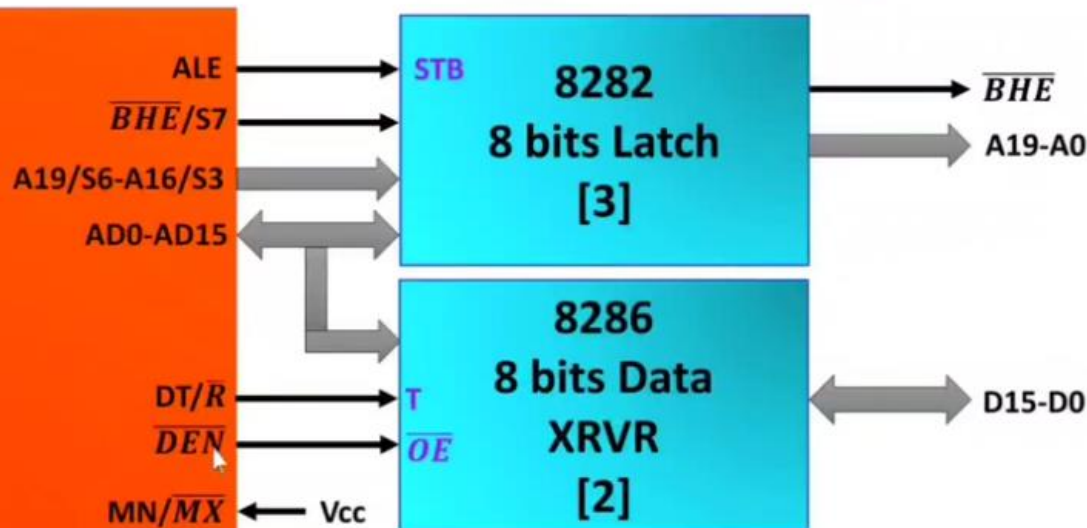
- **T1:** In the first clock cycle of a bus operation, the 8086 outputs the memory address on these lines.
- **T2, T3, T4:** In the subsequent cycles, the same lines are used to transmit or receive data.

The Role of **ALE** (Address Latch Enable):

- **ALE = 1 (High):** the information currently on the AD0-AD15 lines is a valid address (A0-A15). An external latch (like an 8282 or 74LS373 chip) uses this signal to "capture" and hold the address for the entire duration of the bus cycle.
- **ALE = 0 (Low):** The AD0-AD15 lines are now being used for data (D0-D15).



Address Data Demultiplexing of 8086



- ☐ Here, we will do Address Data Demultiplexing in Minimum Mode. So connect MN/ $\overline{MX}$ with Vcc.
- ☐ In Minimum Mode ALE terminal is available for Address Latch.
- ☐ ALE terminal is use to separate Address, Data and status signals. If ALE = 1, Address is available on this lines.

	$\overline{BHE}/S7$	A19/S6-A16/S3	AD15-AD0
ALE = 1	$\overline{BHE}$	A19-A16	A15-A0
ALE = 0	S7	S6-S3	D15-D0

S4	S3	Segment Accessed
0	0	ES
0	1	SS
1	0	CS
1	1	DS

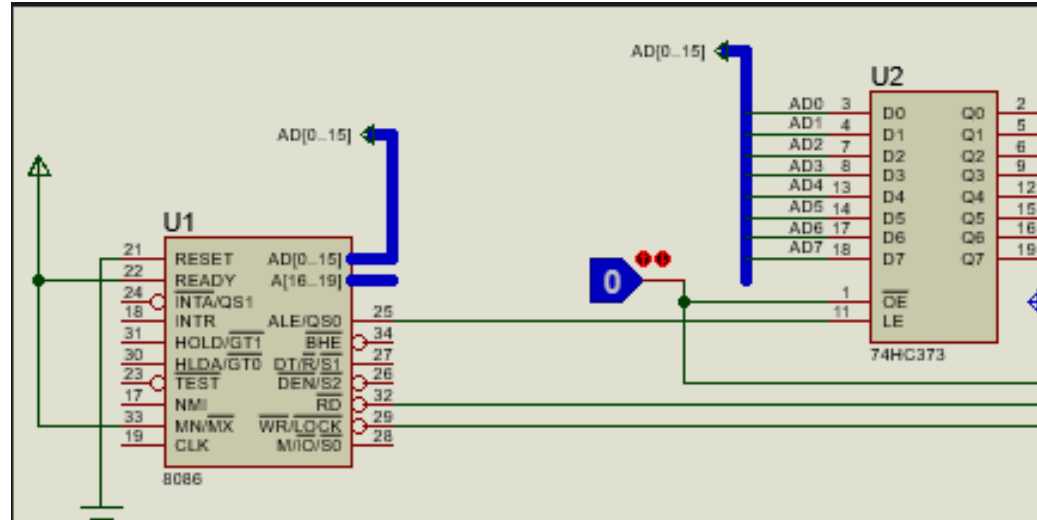
**Microprocessor
8086**

8282 8-bit latch (or 74HC373)

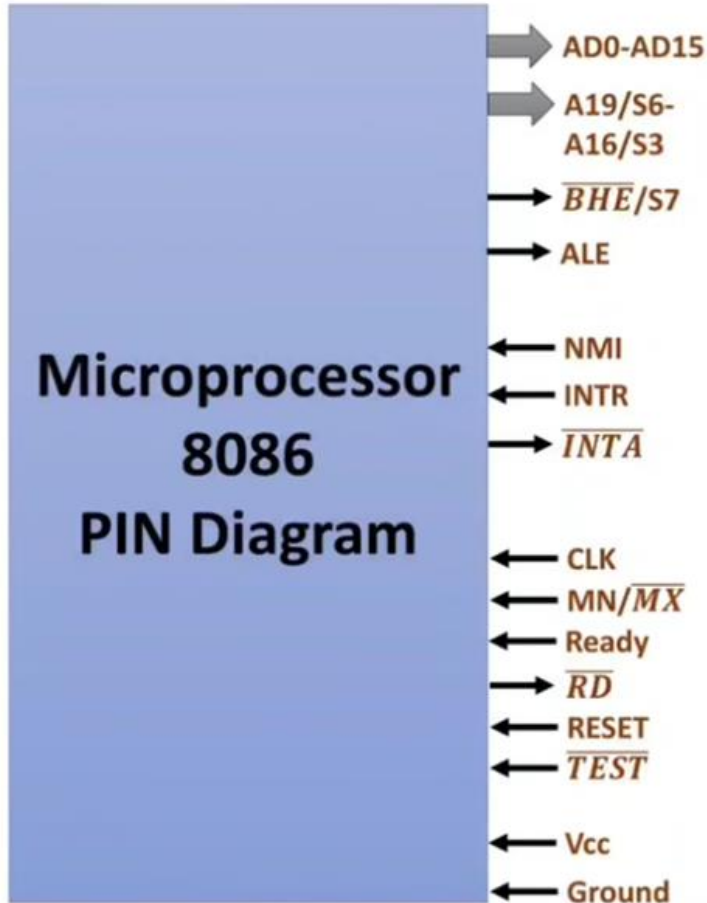
The 8086 uses a **multiplexed address/data bus (AD0–AD15)**, meaning the same lines carry **address** during the first part of the bus cycle and **data** during the second part.

1. During **T1 state of the bus cycle**, the 8086 places the **address** on AD0–AD15.
2. The **ALE (Address Latch Enable)** signal from 8086 goes high, indicating that the address is valid.
3. The **STB** pin of 8282 (**LE** pin 74HC373) is triggered by ALE (usually connected directly).
4. The 8282 **latches the lower byte of the address** (AD0–AD7) and holds it on its outputs (Q0–Q7).
5. The **OE** pin enables the outputs to be used as part of the full address bus.

Note: For AD8–AD15, a second latch (like another 8282 or 74HC373) is used to hold the upper byte of the address.



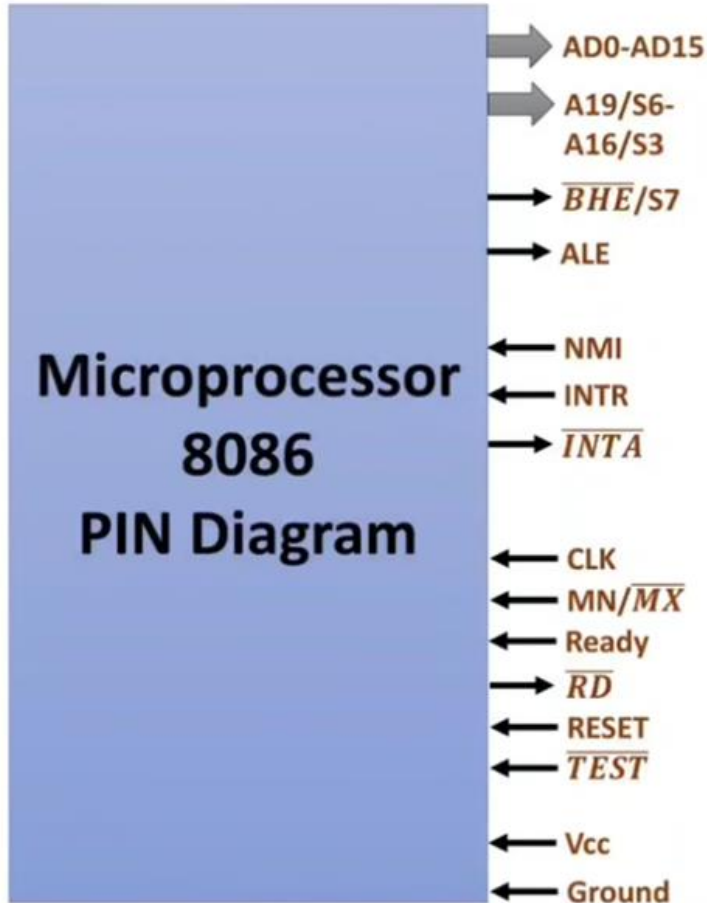
Control Signals in the 8086 Microprocessor



Control signals are essential for managing internal operations and communication with external devices.

These signals determine when data is read, written, and which operations are performed.

Control Signals in the 8086 Microprocessor



Operating Modes

The 8086 can operate in two modes:

- **Minimum Mode:** Used in single-processor systems.
- **Maximum Mode:** Used in multiprocessor systems.

Control signals vary depending on the selected mode.